Real-Time DAQ & Signal Analysis Platform

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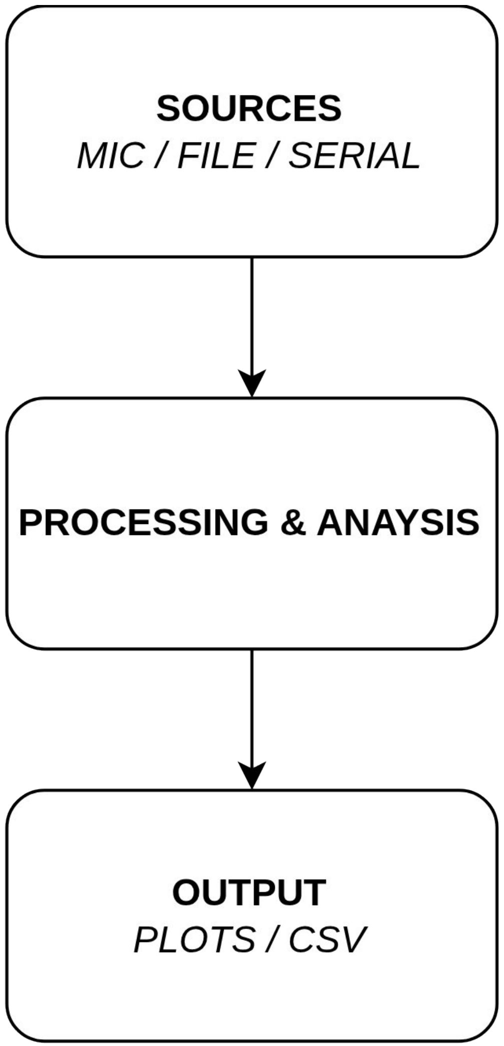
The goal

- **Real time data acquisition and signal analysis**
- **Source agnostic architecture**
- **Signal analysis across time, frequency, and time-frequency domains**
- **Measure a real world engineering problem**



Software architecture

Acquisition

IDataSource
Abstract data source

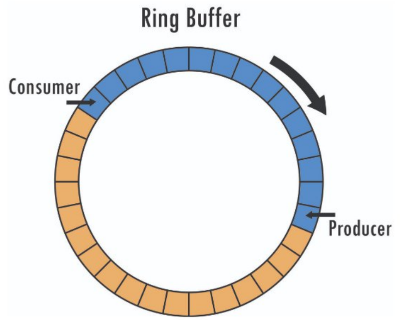
- FileSource — *WAV file support*
- MicrophoneSource — *Microphone support*
- SerialSource — *USB Serial devices*

write(data_block) →

Buffering

RingBuffer
Signal storage

- Thread safe
- Batch writes
- Fixed capacity
- Overwrite oldest on overflow



← read_latest(n)

Processing & visualisation

Main Window
Top level UI

- TimeDomain — *Time domain*
- FrequencyDomain — *Fourier Transform*
- Spectrogram — *STFT*

Control panel

- Plot settings
- DSP parameters
- App metrics
- Export CSV

Language	UI Framework	Signal processing	Data acquisition
	PYQT6	SciPy	sounddevice soundfile pyserial

Real world application

- High school mechatronics diploma project: **coin sorting machine**
- Mechanical switches suffered contact bounce
- What does the bounce signal actually look like?
- Can the DAQ platform analyze it in real time?

Live demo

Prototype trade-offs

Prototype	Production
Python (fast development)	C++ (deterministic latency)
Full recompute every frame	Display decimation
Three sources proved the abstraction	OpenDAQ, additional sensors

THANK YOU FOR THE ATTENTION

QUESTIONS?